



(一) 矩陣

矩陣結構:

Column
↓ $C_1 \ C_2 \ C_3$

row → r_1 $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$
 r_2 $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$...
 r_3 $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$
⋮

1. 單位矩陣 Identity Matrix

$$I_{n \times n} = \begin{bmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \\ 0 & 0 & \dots & 0 & 1 \end{bmatrix}$$

特性: $A \times I = A$
除了中心對角是 1,
其他元素皆為 0。

2. 轉置 Transpose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

特性: $(AB)^T = B^T A^T$

3. Trace

$$\text{tr} \left(\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \right) = 1 + 5 + 9 = 25$$

4. Symmetric Matrix

$$* A = A^T, A = \begin{bmatrix} a & e & f \\ e & b & g \\ f & g & c \end{bmatrix}$$

5. Skew Symmetric Matrix

$$* A = -A^T$$
$$A = \begin{bmatrix} a & e & f \\ -e & b & g \\ -f & -g & c \end{bmatrix}$$

$\swarrow \times -1$

$$A^T = \begin{bmatrix} a & -e & -f \\ e & b & -g \\ f & g & c \end{bmatrix}$$

6. Diagonal Matrix

$$\begin{bmatrix} \circ & \circ & \circ & \circ \\ 0 & & & \\ \circ & & & \\ \circ & & & \end{bmatrix}$$

$$D = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$D \times D \dots D^n = \begin{bmatrix} a^n & 0 & 0 \\ 0 & b^n & 0 \\ 0 & 0 & c^n \end{bmatrix}$$

(=) 矩陣運算

$$\begin{cases} x + y - z = 9 \\ y + 3z = 3 \\ -x - 2z = 2 \end{cases}$$

$$\begin{bmatrix} 1 & 1 & -1 & | & 9 \\ 0 & 1 & 3 & | & 3 \\ -1 & 0 & -2 & | & 2 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 + r_1} \begin{bmatrix} 1 & 1 & -1 & | & 9 \\ 0 & 1 & 3 & | & 3 \\ 0 & 1 & -3 & | & 2 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 - r_2} \begin{bmatrix} 1 & 1 & -1 & | & 9 \\ 0 & 1 & 3 & | & 3 \\ 0 & 0 & -6 & | & 8 \end{bmatrix}$$

$$\xrightarrow{r_3 = -\frac{1}{6}r_3} \begin{bmatrix} 1 & 1 & -1 & | & 9 \\ 0 & 1 & 3 & | & 3 \\ 0 & 0 & -1 & | & -\frac{4}{3} \end{bmatrix}$$

$$\xrightarrow{r_2 = r_2 - 3r_3} \begin{bmatrix} 1 & 1 & -1 & | & 9 \\ 0 & 1 & 0 & | & 7 \\ 0 & 0 & -1 & | & -\frac{4}{3} \end{bmatrix} \xrightarrow{r_1 = r_1 + r_2 + r_3} \begin{bmatrix} 1 & 0 & 0 & | & \frac{2}{3} \\ 0 & 1 & 0 & | & 7 \\ 0 & 0 & -1 & | & -\frac{4}{3} \end{bmatrix}$$

這種形式
叫 "ref" 

7. 8 上下三角

R 上三角 L 下三角

$$\begin{bmatrix} \square & & \\ 0 & \square & \\ 0 & 0 & \square \end{bmatrix}$$

$$\begin{bmatrix} \square & & \\ \square & \square & \\ \square & \square & \square \end{bmatrix}$$

9 增廣矩陣

$$\begin{cases} 3x_1 + 4x_2 = 5 \\ x_1 + 6x_2 = 7 \end{cases} \Rightarrow \begin{bmatrix} 3 & 4 & | & 5 \\ 1 & 6 & | & 7 \end{bmatrix}$$

10. 擴展矩陣

$$* A \rightarrow A^{-1} \Rightarrow A^{-1} [A | I] \xrightarrow{\downarrow} [I | A^{-1}]$$

特殊的例子:

$$(1) \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 6 & -3 & 0 \\ 1 & -2 & -1 & 0 \end{array} \right] \Rightarrow x=0, y=0, z=0$$

This Case is Called: "Trivial Solution"

$$(2) \begin{cases} x_1 + 2x_2 + x_3 = 0 \\ x_1 - x_2 - x_3 = 0 \end{cases} \Rightarrow \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 1 & -1 & 0 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 0 & -\frac{1}{3} \\ 0 & 1 & \frac{2}{3} \end{array} \right] \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{aligned} x_1 - \frac{1}{3}x_3 &= 0 & \Rightarrow x_1 &= \frac{1}{3}x_3 \\ x_2 + \frac{2}{3}x_3 &= 0 & \Rightarrow x_2 &= -\frac{2}{3}x_3 \end{aligned}$$

free variable.

$$\rightarrow \begin{bmatrix} \frac{1}{3}x_3 \\ -\frac{2}{3}x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ 1 \end{bmatrix}$$

此即被稱作
「線性組合」

2 / 25 Problem

$$\begin{cases} 2x_1 + 3x_2 + x_3 + 4x_4 - 9x_5 = 17 \\ x_1 + x_2 + x_3 + x_4 - 3x_5 = 6 \\ x_1 + x_2 + x_3 + 2x_4 - 5x_5 = 8 \\ 2x_1 + 2x_2 + 2x_3 + 3x_4 - 8x_5 = 14 \end{cases}$$

$$\Rightarrow \begin{matrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{matrix} \begin{bmatrix} 2 & 3 & 1 & 4 & -9 & | & 17 \\ 1 & 1 & 1 & 1 & -3 & | & 6 \\ 1 & 1 & 1 & 2 & -5 & | & 8 \\ 2 & 2 & 2 & 3 & -8 & | & 14 \end{bmatrix} \xrightarrow{\text{Swap } (r_1, r_2)} \begin{bmatrix} 1 & 1 & 1 & 1 & -3 & | & 6 \\ 2 & 3 & 1 & 4 & -9 & | & 17 \\ 1 & 1 & 1 & 2 & -5 & | & 8 \\ 2 & 2 & 2 & 3 & -8 & | & 14 \end{bmatrix}$$

$$\begin{matrix} r_2 = r_2 - 2r_1 \\ r_4 = r_4 - 2r_1 \\ r_3 = r_3 - r_1 \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 & -3 & | & 6 \\ 0 & 1 & -1 & 2 & -3 & | & 5 \\ 0 & 0 & 0 & 1 & -2 & | & 2 \\ 0 & 0 & 0 & 1 & -2 & | & 2 \end{bmatrix} \xrightarrow{r_4 = r_4 - r_3} \begin{bmatrix} 1 & 1 & 1 & 1 & -3 & | & 6 \\ 0 & 1 & -1 & 2 & -3 & | & 5 \\ 0 & 0 & 0 & 1 & -2 & | & 2 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{matrix} r_1 = r_1 - r_3 \\ r_2 = r_2 - 2r_3 \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 0 & -1 & | & 4 \\ 0 & 1 & -1 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & 1 & -2 & | & 2 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix} \xrightarrow{r_1 = r_1 - r_2} \begin{bmatrix} 1 & 0 & 2 & 0 & -2 & | & 3 \\ 0 & 1 & -1 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & 1 & -2 & | & 2 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

列成增廣矩陣並求其簡化梯形矩陣後，
求其線性組合。

*若一矩陣B可由矩陣A經一些基本運算所得，
則稱A與B為列等價 (Row Equivalent)

*簡化梯形矩陣 — Reduced Row-Echelon Form)

— A的非零列之第1個非零元素為1，稱其為Leading 1.

— 含Leading 1的行 (Column) 內，其他元素為0

— 所有Leading 1由右上左下分佈，即 a_{ij} 和 a_{mn} 為Leading 1，有 $(m-i)(n-j) > 0$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 3 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \text{ 求線性組合。}$$

$$\rightarrow \begin{cases} x_1 + 2x_3 - 2x_5 = 3 \\ x_2 - x_3 + x_5 = 1 \\ x_4 - 2x_5 = 2 \end{cases}$$

$$\rightarrow \begin{cases} x_1 = 3 - 2x_3 + 2x_5 \\ x_2 = 1 + x_3 - x_5 \\ x_4 = 2 + 2x_5 \end{cases} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 3 - 2x_3 + 2x_5 \\ 1 + x_3 - x_5 \\ x_3 \\ 2 + x_3 + 2x_5 \\ x_5 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 2 \\ -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} \#$$

(二) 矩陣運算

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

$$A - B = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

$$A \times B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

- 轉置の利用:

$$A_{2 \times 3} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, B_{1 \times 3} = [1, 2, 1], B_{3 \times 1}^T = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$A \times B^T = \begin{bmatrix} 8 \\ 20 \end{bmatrix}$$

- 旋轉矩陣

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \text{ or } \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

逆時針 順時針

$$A^2 = AA = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A^3 = AAA$$

$$B^k = \begin{bmatrix} (2)^k & 0 & 0 \\ 0 & (-3)^k & 0 \\ 0 & 0 & (5)^k \end{bmatrix}$$

e.g.

$$A_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$B_{3 \times 1} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

3/4 Problem

$$1. A = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \sqrt{\frac{3}{4}} \end{bmatrix}, A^6 = ?$$

$$\cos \theta = \frac{2}{2\sqrt{3}}, \theta = 30^\circ \quad \sin \theta = \frac{1}{2}, \theta = 30^\circ$$

$$A = \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{bmatrix}, A^6 = \begin{bmatrix} \cos 180^\circ & -\sin 180^\circ \\ \sin 180^\circ & \cos 180^\circ \end{bmatrix} \\ = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}^*$$

$$2. A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}, B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{bmatrix}$$

$$3. \sum_{i=0}^n \begin{bmatrix} 1 & i \\ i^2 & i^3 \end{bmatrix}$$

$$\sum_{i=0}^n \begin{bmatrix} 1 & i \\ i^2 & i^3 \end{bmatrix} = \begin{bmatrix} n+1 & \frac{n(n+1)}{2} \\ \frac{n(n+1)(2n+1)}{6} & \left[\frac{n(n+1)}{2} \right]^2 \end{bmatrix}$$

Leading 1 个数表示法: $\text{rank}(A)$

e.g. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A, \text{rank}(A) = 3$

$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, find $\text{rank}(A)$

$\xrightarrow{L} \text{rref}(A) = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{rank}(A) = 2$

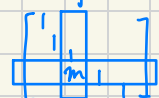
初等矩阵 Elementary Matrix.
(基本矩阵)

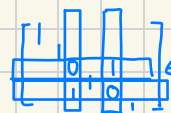
$[E_1] \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \times E_1 = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & 6 \\ 7 & 8 & 9 \end{bmatrix}$
($k_2 = k_2 - 4k_1$)

Formula:

1. $r_i = m r_i = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \leftarrow i$

2. $r_i = r_i + m r_j$ 

3. Swap(r_i, r_j) 

注意 基本矩阵的
建構 & 计算

The Calculate's ways of Elementary Matrix.

以 $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 4 \end{bmatrix}$ 来看。

$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 4 \end{bmatrix}, E_1, E_2$

$\xrightarrow{r_3 = r_3 - r_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 0 & 3 & -1 \end{bmatrix} \xrightarrow{E_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

$\xrightarrow{r_2 = -\frac{1}{2}r_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & -3 & 1 \end{bmatrix} \xrightarrow{E_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\xrightarrow{r_3 = r_3 - r_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 0 & 3 & -1 \end{bmatrix} \xrightarrow{r_2 = -\frac{1}{2}r_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & -3 & 1 \end{bmatrix}$

E_1 Matrix

Build Steps:

$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\xrightarrow{r_3 = r_3 - r_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

E_2 Matrix

Build Steps:

$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\xrightarrow{r_2 = -\frac{1}{2}r_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

反矩陣 $A^{-1} \cdot A = I$

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{bmatrix}, A^{-1}?$$

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{bmatrix} \xrightarrow{\substack{x E_1 \downarrow \\ x E_2 \downarrow \\ \vdots \\ x E_k \downarrow \\ \text{rref}}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad [E_k \dots E_2 E_1][A] = I$$

可逆性 $BA = AB = I$

1. 矩陣運算; 2. 線性變換

$$[A | I] = [I | A^{-1}]$$

$$\text{let } A = \begin{bmatrix} 5 & 1 & -2 \\ -2 & 0 & 5 \\ 7 & 2 & 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 5 & 1 & -2 & 1 & 0 & 0 \\ -2 & 0 & 5 & 0 & 1 & 0 \\ 7 & 2 & 0 & 0 & 0 & 1 \end{array} \right] \rightarrow A^{-1}$$

$$\xrightarrow{\text{rref?}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & ? & ? & ? \\ 0 & 1 & 0 & ? & ? & ? \\ 0 & 0 & 1 & ? & ? & ? \end{array} \right]$$

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 - r_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 0 & -3 & -2 \end{bmatrix}, E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_2 = -\frac{1}{2}r_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & -3 & -2 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 + 3r_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

$$\xrightarrow{r_2 = r_2 - r_3} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_1 = r_1 - 3r_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_5 = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ 因此, A^{-1} 存在 >

$$[E_5 E_4 E_3 E_2 E_1][A] = I$$

A^{-1}

反矩陣公式解:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$B^{-1} = \frac{1}{\det(B)} \begin{bmatrix} |e & f| & -|bc| & |bc| \\ -|d & f| & |ac| & -|ac| \\ |de| & |ab| & |ab| & |de| \end{bmatrix}$$

3/11 Problem

1. $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{bmatrix}$, find $\text{rank}(A)$.

Back >^c

2. $A = \begin{bmatrix} 1 & 0 & -8 \\ -4 & 9 & 0 \\ 0 & -9 & 1 \end{bmatrix}$, find all elementary matrix.

$$\begin{bmatrix} 1 & 0 & -8 \\ -4 & 9 & 0 \\ 0 & -9 & 1 \end{bmatrix}$$

$$\xrightarrow{r_2 = r_2 + 4r_1} \begin{bmatrix} 1 & 0 & -8 \\ 0 & 9 & -32 \\ 0 & -9 & 1 \end{bmatrix}, E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 + r_2} \begin{bmatrix} 1 & 0 & -8 \\ 0 & 9 & -32 \\ 0 & 0 & -31 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow{r_2 = \frac{1}{9}r_2} \begin{bmatrix} 1 & 0 & -8 \\ 0 & 1 & -\frac{32}{9} \\ 0 & 0 & -31 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_3 = -\frac{1}{31}r_3} \begin{bmatrix} 1 & 0 & -8 \\ 0 & 1 & -\frac{32}{9} \\ 0 & 0 & 1 \end{bmatrix}, E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{31} \end{bmatrix}$$

$$\xrightarrow{r_2 = r_2 + \frac{32}{9}r_3} \begin{bmatrix} 1 & 0 & -8 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{32}{9} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_1 = r_1 + 8r_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_6 = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \neq$$

3. $\begin{bmatrix} 5 & 1 & -2 \\ -2 & 0 & 5 \\ 7 & 2 & 8 \end{bmatrix}$, find A^{-1}

$$\left[\begin{array}{ccc|ccc} 5 & 1 & -2 & 1 & 0 & 0 \\ -2 & 0 & 5 & 0 & 1 & 0 \\ 7 & 2 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$r_3 = r_3 - 2r_1 \rightarrow \left[\begin{array}{ccc|ccc} 5 & 1 & -2 & 1 & 0 & 0 \\ -2 & 0 & 5 & 0 & 1 & 0 \\ -3 & 0 & 12 & -2 & 0 & 1 \end{array} \right]$$

$$r_3 = \frac{1}{3}r_3 \rightarrow \left[\begin{array}{ccc|ccc} 5 & 1 & -2 & 1 & 0 & 0 \\ -2 & 0 & 5 & 0 & 1 & 0 \\ -1 & 0 & 4 & -\frac{2}{3} & 0 & \frac{1}{3} \end{array} \right]$$

$$r_1 = r_1 + 5r_3 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 18 & -\frac{7}{3} & 0 & \frac{5}{3} \\ -2 & 0 & 5 & 0 & 1 & 0 \\ -1 & 0 & 4 & -\frac{2}{3} & 0 & \frac{1}{3} \end{array} \right]$$

$$r_2 = r_2 - 2r_3 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 18 & -\frac{7}{3} & 0 & \frac{5}{3} \\ 0 & 0 & -3 & \frac{4}{3} & 1 & -\frac{2}{3} \\ -1 & 0 & 4 & -\frac{2}{3} & 0 & \frac{1}{3} \end{array} \right]$$

$$r_2 = -\frac{1}{3}r_2 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 18 & -\frac{7}{3} & 0 & \frac{5}{3} \\ 0 & 0 & 1 & -\frac{4}{9} & -\frac{1}{3} & \frac{2}{9} \\ -1 & 0 & 4 & -\frac{2}{3} & 0 & \frac{1}{3} \end{array} \right]$$

$$r_1 = r_1 - 18r_2 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & \frac{17}{3} & 6 & -\frac{7}{3} \\ 0 & 0 & 1 & -\frac{4}{9} & -\frac{1}{3} & \frac{2}{9} \\ -1 & 0 & 4 & -\frac{2}{3} & 0 & \frac{1}{3} \end{array} \right]$$

$$r_3 = r_3 - 4r_2 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & \frac{17}{3} & 6 & -\frac{7}{3} \\ 0 & 0 & 1 & -\frac{4}{9} & -\frac{1}{3} & \frac{2}{9} \\ -1 & 0 & 0 & \frac{10}{9} & \frac{4}{3} & -\frac{5}{9} \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 0 & \frac{17}{3} & 6 & -\frac{7}{3} \\ 0 & 0 & 1 & -\frac{4}{9} & -\frac{1}{3} & \frac{2}{9} \\ -1 & 0 & 0 & \frac{10}{9} & \frac{4}{3} & -\frac{5}{9} \end{array} \right]$$

$$r_3 = -r_3 \rightarrow \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & \frac{17}{3} & 6 & -\frac{7}{3} \\ 0 & 0 & 1 & -\frac{4}{9} & -\frac{1}{3} & \frac{2}{9} \\ 1 & 0 & 0 & -\frac{10}{9} & -\frac{4}{3} & \frac{5}{9} \end{array} \right]$$

$$\text{Swap}(r_3, (r_1, r_2)) \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{10}{9} & -\frac{4}{3} & \frac{5}{9} \\ 0 & 1 & 0 & \frac{17}{3} & 6 & -\frac{7}{3} \\ 0 & 0 & 1 & -\frac{4}{9} & -\frac{1}{3} & \frac{2}{9} \end{array} \right]$$

1. $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{bmatrix}$, 求 $\text{rank}(A)$ 。

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{bmatrix} \xrightarrow{r_3 = r_3 - r_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -2 & -2 \\ 1 & -3 & 1 \end{bmatrix}$$

$$\xrightarrow{r_2 = r_2 + r_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 1 & -3 & 1 \end{bmatrix}$$

$$\text{rank}(A) = 3$$

$$\xrightarrow{r_3 = r_3 + 2r_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 - r_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\xrightarrow{r_3 = \frac{1}{2}r_3} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_1 = r_1 - 3r_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$A \times B$ 如何加速?

矩陣分割!

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \cancel{A_1 B_1} + \cancel{A_2 B_3} & \cancel{A_1 B_2} + \cancel{A_2 B_4} \\ \cancel{A_3 B_1} + \cancel{A_4 B_3} & \cancel{A_3 B_2} + \cancel{A_4 B_4} \end{bmatrix} = \begin{bmatrix} B_1 & B_4 \\ 0 & A_4 B_4 \end{bmatrix}$$

$$A_4 B_4 = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ -7 & 1 \end{bmatrix}$$

$$\Downarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ -1 & 1 & -1 & 1 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & -7 & 1 \end{bmatrix}$$

怎麼切？

！請記得左邊直線跟右邊的橫線需要對應。

— LU 分解

$$A = LU$$

L: 下三角矩陣 $\begin{bmatrix} 1 & 0 & 0 \\ \times & 1 & 0 \\ \times & \times & 1 \end{bmatrix}$

U: 上三角矩陣 $\begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \end{bmatrix}$

$$\begin{bmatrix} 2 & 2 & 3 \\ 5 & 9 & 10 \\ 4 & 1 & 2 \end{bmatrix} \xrightarrow[r_3 = r_3 - 2r_2]{r_2 = r_2 - \frac{5}{2}r_1} \begin{bmatrix} 2 & 2 & 3 \\ 0 & 4 & \frac{5}{2} \\ 0 & -3 & 4 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 + \frac{3}{4}r_2} \begin{bmatrix} 2 & 2 & 3 \\ 0 & 4 & \frac{5}{2} \\ 0 & 0 & -\frac{17}{8} \end{bmatrix} = U$$

$$L = \begin{matrix} & L_1 & L_2 & L_3 \\ \begin{matrix} r_1 \\ r_2 \\ r_3 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 \\ \frac{5}{2} & 1 & 0 \\ 2 & -\frac{3}{4} & 1 \end{bmatrix} \end{matrix}$$

⚠ 請記得，LU 分解只能做三次！

求 L 記得變號！

3/18 Problem

$$1. \begin{bmatrix} A_1 & A_2 & A_3 \\ 1 & 2 & 4 & 8 & 5 & 5 \\ 9 & -1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 5 & 3 & 3 & 3 \\ 4 & 10 & 5 & 8 & 8 & 8 \\ A_4 & A_5 & A_6 \end{bmatrix} \begin{matrix} B_1 & B_2 \\ \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 10 \\ 0 & 0 & 0 & 2 & 0 \\ 1 & 1 & 4 & 1 & 1 \\ -1 & -1 & -4 & -1 & 1 \end{bmatrix} \\ B_3 & B_4 \\ B_5 & B_6 \end{matrix}$$

$$\begin{matrix} \textcircled{1} \\ \textcircled{3} \end{matrix} \begin{bmatrix} A_1 B_1 + \cancel{A_2 B_3} + A_3 B_5 & \cancel{A_1 B_2} + A_2 B_4 + A_3 B_6 \\ A_4 B_1 + \cancel{A_5 B_2} + A_6 B_5 & \cancel{A_4 B_2} + A_5 B_4 + A_6 B_6 \end{bmatrix} \begin{matrix} \textcircled{2} \\ \textcircled{4} \end{matrix}$$

$$A_1 B_1 = \begin{bmatrix} 1 & 2 \\ 9 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 18 & -2 \end{bmatrix} \quad A_3 B_5 = \begin{bmatrix} 5 & 5 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\textcircled{1} \begin{bmatrix} 2 & 4 \\ 18 & -2 \end{bmatrix}$$

$$A_2 B_4 = \begin{bmatrix} 4 & 8 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 10 \\ 0 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 8 & 16 & 40 \\ 0 & 0 & 0 \end{bmatrix} \quad \textcircled{2} \begin{bmatrix} 8 & 16 & 40 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 10 \\ 0 & 0 & -2 \end{bmatrix}$$

$$A_3 B_6 = \begin{bmatrix} 5 & 5 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 \\ -4 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 10 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 8 & 16 & 50 \\ 0 & 0 & -2 \end{bmatrix}$$

$$1. \begin{bmatrix} A_1 & A_2 & A_3 \\ 1 & 2 & 4 & 8 & 55 \\ 9 & -1 & 0 & 0 & -1 \\ 2 & 2 & 5 & 3 & 3 \\ 4 & 10 & 5 & 8 & 8 \\ A_4 & A_5 & A_6 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ 2 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 10 \\ 0 & 0 & 0 & 2 & 0 \\ 1 & 1 & 4 & 1 & 1 \\ -1 & -1 & -4 & -1 & 1 \\ B_3 & B_4 & B_5 & B_6 \end{bmatrix}$$

$$\textcircled{1} \begin{bmatrix} A_1 B_1 + A_2 B_3 + A_3 B_5 & A_1 B_2 + A_2 B_4 + A_3 B_6 \end{bmatrix} \textcircled{2}$$

$$\textcircled{3} \begin{bmatrix} A_4 B_1 + A_5 B_2 + A_6 B_5 & A_4 B_2 + A_5 B_4 + A_6 B_6 \end{bmatrix} \textcircled{4}$$

$$A_4 B_1 = \begin{bmatrix} 2 & 2 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 8 & 20 \end{bmatrix} \textcircled{3} \begin{bmatrix} 4 & 4 \\ 8 & 20 \end{bmatrix}$$

$$A_6 B_5 = \begin{bmatrix} 3 & 3 \\ 8 & 8 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A_5 B_4 = \begin{bmatrix} 5 & 3 \\ 5 & 8 \end{bmatrix} \begin{bmatrix} 20 & 10 \\ 0 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 10 & 6 & 50 \\ 10 & 16 & 50 \end{bmatrix} \textcircled{4} \begin{bmatrix} 10 & 6 & 56 \\ 10 & 16 & 66 \end{bmatrix}$$

$$A_6 B_6 = \begin{bmatrix} 3 & 3 \\ 8 & 8 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 \\ -4 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 16 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 8 & 16 & 50 \\ 18 & -2 & 0 & 0 & -2 \\ 4 & 4 & 10 & 6 & 56 \\ 8 & 20 & 10 & 16 & 66 \end{bmatrix}$$

$$2. \begin{bmatrix} 2 & 8 & 6 \\ -8 & -22 & -20 \\ 10 & 0 & 3 \end{bmatrix} \text{ LU 分解}$$

U:

$$\begin{bmatrix} 2 & 8 & 6 \\ -8 & -22 & -20 \\ 10 & 0 & 3 \end{bmatrix} \xrightarrow{r_2 = r_2 + 4r_1} \begin{bmatrix} 2 & 8 & 6 \\ 0 & 10 & 4 \\ 10 & 0 & 3 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 - 5r_1} \begin{bmatrix} 2 & 8 & 6 \\ 0 & 10 & 4 \\ 0 & -40 & -27 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 + 4r_2} \begin{bmatrix} 2 & 8 & 6 \\ 0 & 10 & 4 \\ 0 & 0 & -11 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 5 & -4 & 1 \end{bmatrix} \times$$

04/15

線性獨立 V.S. 線性相依
Linear Independent Linear Dependent

$$S = \{V_1, V_2, \dots, V_n\}$$

存在 $V_i = V_a, V_b, V_c \dots$

$$xV_a + xV_b + xV_c = V_i$$

→ Conclusion: 假設 V_i 能夠被 V_1, V_2, V_3 組合出來,
則稱 V_i 線性相依, 否則就是線性獨立。

線性相依的情況

$$S = \{V_1, V_2, V_3, \dots, V_n\}$$

(1)

$$\begin{bmatrix} V_1 & V_2 & V_3 & \dots & V_n \end{bmatrix}$$

$$x_1 V_1 + x_2 V_2 + \dots + x_n V_n = 0$$

(2)

$$\begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix} \leftarrow r_i$$

⇒ 則稱線性相依。

↳ rref → $\begin{bmatrix} 0 \end{bmatrix}$ ← 其中有一列為 0。

ex

$$S = \left\{ \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -4 \\ 6 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\} \text{ independent / dependent?}$$

$$\begin{bmatrix} 2 & 2 & 1 \\ -4 & 6 & 5 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow{\text{swp}(r_1, r_2)} \begin{bmatrix} 1 & 0 & 0 \\ -4 & 6 & 5 \\ 2 & 2 & 1 \end{bmatrix} \xrightarrow{\substack{r_2 = r_2 + 4r_1 \\ r_3 = r_3 - 2r_1}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 6 & 5 \\ 0 & 2 & 1 \end{bmatrix} \xrightarrow{\text{swp}(r_2, r_3)} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 6 & 5 \end{bmatrix} \xrightarrow{r_2 = r_2 - 3r_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \xrightarrow{r_3 = \frac{1}{2}r_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = \left\{ \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -4 \\ 6 \\ 5 \end{pmatrix}, \begin{pmatrix} -2 \\ 8 \\ 6 \end{pmatrix} \right\} \text{ independent / dependent?}$$

$$\begin{matrix} r_2 = r_2 - r_3 \\ r_2 = \frac{1}{2}r_2 \end{matrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 & 1 \\ -4 & 6 & 5 \\ -2 & 8 & 6 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 2 & 0 & -\frac{3}{5} \\ 0 & 1 & \frac{7}{10} \\ 0 & 0 & 0 \end{bmatrix}$$

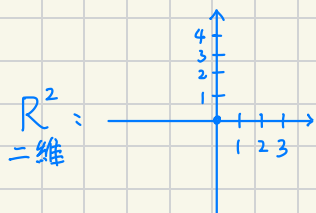
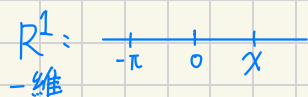
$r_3 = 0 \rightarrow$ 線性相依!

線性獨立 ✗

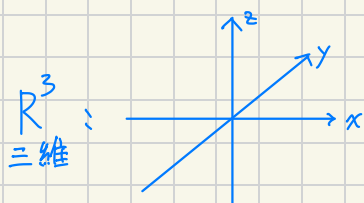
R 空間

基底

(可透過此向量來生成表示 R 空間的所有位置)



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$



$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$\text{Span}\{\nu_1, \nu_2, \dots, \nu_n\} = \nu_1, \nu_2, \dots, \nu_n$ 可以組合出來的向量集合。

$$a_1\nu_1 + a_2\nu_2 + \dots + a_n\nu_n = \text{span}\{\nu_1, \nu_2, \dots, \nu_n\}, \forall a_i, 1 \leq i \leq n.$$

ex.

$\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ 在 $\text{Span}\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}\}$ 中嗎?

$$\Rightarrow a_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \text{ 若 } a_1, a_2 \text{ 有解, 則成立。}$$

$$\left[\begin{array}{cc|c} 1 & -1 & 2 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \end{array} \right] \xrightarrow{r_3 = r_3 - 2r_1} \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 3 & -3 \end{array} \right], \therefore \text{可得 } a_2 = -1, a_1 = 1, \text{ 所以 } \begin{pmatrix} 2 \\ -1 \end{pmatrix} \text{ 在此 Span 中。}$$

$n=3, \text{span}\{\nu_1, \nu_2, \nu_3\} = R^n = R^3$?

只要 ν_1, ν_2, ν_3 可能是 $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, 則 $R^n = R^3$!

$$\Rightarrow \nu_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \text{span}\{\nu_1, \nu_2, \nu_3\} = R^3?$$

$$\begin{array}{c} \nu_1 \nu_2 \nu_3 \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{ref}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \Rightarrow \text{Span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\} = \text{Span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\} \times$$

$$\nu_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

一个多项式是否能被其他多项式组合?

$$2x^3 - x^2 + x + 3 \in \text{Span}\left(\left\{\begin{bmatrix} x^3 + x^2 + x + 1 \\ x^2 + x + 1 \\ x + 1 \end{bmatrix}\right\}\right)?$$

$$\begin{pmatrix} ? \\ ? \\ ? \\ ? \end{pmatrix} = \begin{pmatrix} x^3 \\ x^2 \\ x \\ 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 1 \\ 3 \end{pmatrix} \in \text{Span}\left(\left\{\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}\right\}\right)$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 1 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 3 \end{array} \right] \begin{array}{l} r_2 = r_2 - r_1 \\ r_3 = r_3 - r_1 \\ r_4 = r_4 - r_1 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 1 \end{array} \right]$$

無解!

$$\therefore 2x^3 - x^2 + x + 3 \notin \text{Span}\left(\left\{\begin{bmatrix} x^3 + x^2 + x + 1 \\ x^2 + x + 1 \\ x + 1 \end{bmatrix}\right\}\right)$$

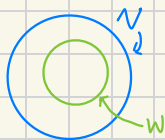
向量空間 Vector Space

$x, y \rightarrow$ 向量, $a, b \rightarrow$ 係數

1. 交換律: $x + y = y + x$
2. 交換律: $x + (y + z) = (x + y) + z$
3. 存在 0 向量: $x + 0 = x$
4. $x + y = 0$
5. $x \cdot 1 = x$
6. $ax = a(bx)$
7. $a(x + y) = ax + ay$
8. $(a + b)x = ax + bx$

子空間 Sub Space

$$W \subseteq V$$



證 W 為 Vector Space?

1. 0 存在於 W

$$2. x + y \in W$$

$$3. cx \in W$$

$$\xrightarrow{\text{ex.}} x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, c = 7$$

$$cx = \begin{pmatrix} 7 \\ 0 \\ 0 \end{pmatrix} \in \mathbb{R}^3 = W$$

$\xrightarrow{\text{ex.}}$

$$W = \mathbb{R}^3$$

$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\therefore x + y = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \in \mathbb{R}^3 = W$$

$\therefore$ PASS!

$U = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, x \leq y \leq z, U$ 是 Vector Space 嗎?

① 0 是否存在 U 中

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \because \text{let } x=0, y=0, z=0 \\ \text{滿足 } x \leq y \leq z$$

$\therefore$ Pass!

② $x+y \in W$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} x_1+x_2 \\ y_1+y_2 \\ z_1+z_2 \end{pmatrix}$$

$$\rightarrow \because x_1+x_2 \leq y_1+y_2 \leq z_1+z_2$$

Pass!

$$\because x_1 \leq y_1 \leq z_1, x_2 \leq y_2 \leq z_2$$

ex.

$p(x) = ax^3, a \in \mathbb{R}, p(x)$ 是 P_3 的子空間嗎?
(Polynomial)

1. $0 \in P_3$?

$a=0$ or $x=0$ 時, $p(x)=0$, Pass!

2. $x+y \in P_3$?

$$\begin{matrix} \uparrow & \nwarrow & \nearrow \\ a_1x^3 & a_2x^3 & ax^3 \end{matrix}$$

$$a_1x^3 + a_2x^3 = \underbrace{(a_1+a_2)}_{=a}x^3 \in P_3 \Rightarrow \text{Pass!}$$

3. $cx \in P_3$

$$\downarrow \\ ax^3$$

$$\downarrow$$

$$ca_1x^3 \leftarrow a_1x^3 \in P_3 \Rightarrow \text{Pass!}$$

$\therefore$ 三項檢定都得證, 所以 $p(x)$ 是 P^3 的子空間。

③ $cx \in U$

$$c \in \mathbb{R} \quad \text{let } c=-1 \quad \begin{pmatrix} 1 \\ 5 \\ 10 \end{pmatrix} \rightarrow cx = \begin{pmatrix} -1 \\ -5 \\ -10 \end{pmatrix} \rightarrow x > y > z \text{ 不滿足!} \\ \text{所以 } \begin{pmatrix} -1 \\ -5 \\ -10 \end{pmatrix} \text{ 不是 Vector Space}$$



假设題目問:

U 是否是 $\mathbb{R}^3$ 的子空間 (SubSpace)

其等價於本題的「 U 是否是 Vector Space」?

04/15 Problem

1. $A_1 = (4x^2 - 4x + 7)$ $S = \{A_1, A_2, A_3\}$
 $A_2 = (-3x^2 + 2x + 6)$
 $A_3 = (5x^2 - 6x + 20)$ ' S is independent / dependent?

$$\begin{pmatrix} 4 \\ -4 \\ 7 \end{pmatrix}, \begin{pmatrix} -3 \\ 2 \\ 6 \end{pmatrix}, \begin{pmatrix} 5 \\ -6 \\ 20 \end{pmatrix}$$

$$\begin{bmatrix} 4 & -4 & 7 \\ -3 & 2 & 6 \\ 5 & -6 & 20 \end{bmatrix} \xrightarrow{\substack{r_1 = r_1 + r_2 \\ r_3 = r_3 + r_2}} \begin{bmatrix} 1 & -2 & 13 \\ -3 & 2 & 6 \\ 2 & -4 & 26 \end{bmatrix} \xrightarrow{\substack{r_2 = r_2 + 3r_1 \\ r_3 = r_3 - 2r_1}} \begin{bmatrix} 1 & -2 & 13 \\ 0 & -4 & 45 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{Dependent 相依}$$

2. $A_1 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$
 $A_2 = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$ $S = \{A_1, A_2, A_3\}$
 $A_3 = \begin{pmatrix} 3 & -2 \\ 1 & 2 \end{pmatrix}$ ' S is independent / dependent?

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & -1 & 1 & 0 \\ 3 & -2 & 1 & 2 \end{bmatrix} \xrightarrow{\substack{r_2 = r_2 - 2r_1 \\ r_3 = r_3 - 3r_1}} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -5 & -5 & -8 \\ 0 & -8 & -8 & -10 \end{bmatrix} \xrightarrow{\substack{r_2 = -r_2 \\ r_3 = -\frac{r_3}{2}}} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 5 & 8 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow{r_2 = \frac{r_2}{5}} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & \frac{8}{5} \\ 0 & 1 & 1 & 2 \end{bmatrix} \rightarrow \text{independent 獨立。}$$

3. $\text{span}\left\{\begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}\right\} \in \mathbb{R}_3$?

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 3 & 4 & 5 \end{bmatrix} \xrightarrow{\substack{r_2 = r_2 - 2r_1 \\ r_3 = r_3 - 3r_1}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -2 & -4 \end{bmatrix} \xrightarrow{\substack{r_2 = -\frac{r_2}{3} \\ r_2 = -\frac{r_2}{3}}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{r_3 = r_3 - r_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \notin \mathbb{R}_3$$

4. $\begin{pmatrix} 4 \\ 5 \\ 23 \end{pmatrix}$ 在 $\text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}\right\}$ 中嗎?

✓ $a_1 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + a_2 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 23 \end{pmatrix}$

$$\left[\begin{array}{cc|c} 1 & -1 & 4 \\ 0 & 1 & 5 \\ 2 & 1 & 23 \end{array} \right] \xrightarrow{r_3 = r_1 + r_2} \left[\begin{array}{cc|c} 1 & 0 & 9 \\ 0 & 1 & 5 \\ 2 & 1 & 23 \end{array} \right], a_1 = 9, a_2 = 5, \text{ 存在於 } \text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}\right\} \text{ 中。}$$

5. $V = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, x, y, z \in \mathbb{R}$ 且 $x = y = z$, V 是 $\mathbb{R}^3$ 的 SubSpace 嗎?

6. $V = \begin{pmatrix} 3+a+2b \\ 4+a+3b \\ 5+a+4b \end{pmatrix}, a, b \in \mathbb{R}$, V 是 $\mathbb{R}^3$ 的子空間嗎?

4/22

V 是一個向量空間 (Vector Space), $W \subseteq V$

則滿足:

① $0 \in W$

② $x + y \in W$

③ 常數 乘法封閉性, $c x \in W$

判斷子空間 Another Way

$v_1, v_2, v_3, \dots, v_n \in V$

$\text{span}\{v_1, v_2, \dots, v_n\} \subseteq V$

$U = \begin{bmatrix} a+2b+c \\ 0 \\ 0 \end{bmatrix}, a, b, c \in \mathbb{R}$, U 是否為 $\mathbb{R}^3$ 的子空間 (subspace)

$U = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right\}$

$\in \mathbb{R}^3$, 所以 U 是 $\mathbb{R}^3$ 的 subspace.

$U = \begin{bmatrix} 1+b+c \\ 4+2b+c \\ 5+3b+2c \end{bmatrix}, b, c \in \mathbb{R}$

$U = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix} + b \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

c 是常數向量, 不能直接 span!

看看後面的 b, c 是否可以組合成 $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

$\therefore \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix} \in \text{span}\left\{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}\right\}$

($b = -3, c = 2$ 可組合出) $\therefore$ 所以 U 是 $\mathbb{R}^3$ 的 subspace.

$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} \right\}$ 能構成 $\mathbb{R}^3$ 空間的基底?

若三向量互相獨立, 則可以構成 $\mathbb{R}^3$ 空間的基底。

(做 ref 結果為單位矩陣)

$\begin{bmatrix} -1 & 2 & 0 \\ 0 & 1 & 3 \\ 1 & 4 & 1 \end{bmatrix} \xrightarrow{\text{ref}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ independent! $\rightarrow$ 可以 $\triangleright \triangleleft$

$$A = \begin{bmatrix} 1 & 4 & 8 & -1 \\ 2 & 4 & 8 & 0 \\ 3 & 2 & 4 & 1 \end{bmatrix} \text{ 求 } \text{col}(A) \text{ 的基底 (basis)} \rightarrow \text{每一个 col 當一个向量}$$

$$\rightarrow \text{span}\left\{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 8 \\ 8 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\right\} = \text{col}(A)$$

$$\begin{bmatrix} 1 & 4 & 8 & -1 \\ 2 & 4 & 8 & 0 \\ 3 & 2 & 4 & 1 \end{bmatrix} \xrightarrow{\text{ref}} \begin{bmatrix} 1 & 4 & 8 & -1 \\ 0 & 0 & 2 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow \text{col}(A) \text{ 的 basis} = \left\{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\right\} \Rightarrow \mathbb{R}^3 \text{ 空間的 basis!}$$

不是 ref! pivot!

SPECIAL Problem:

擴充 $\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right\}$ 為 $\mathbb{R}^3$ 的 Basis

$$\begin{bmatrix} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{ref}} \begin{bmatrix} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 1 & 1 \end{bmatrix} \Rightarrow \left\{\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$$

→ 選最左因(越左越好)的 1

零空間 (null)

$$A = \begin{bmatrix} -2 & -9 & -1 \\ 2 & -6 & 5 \\ -4 & -3 & 4 \end{bmatrix}, x = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, x \in \text{null}(A)? \text{ (或寫 } \ker(A) \text{)}$$

假設 x 是 A 的零空間, 則 $Ax = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (0 向量)

$$\begin{bmatrix} -13 \\ -14 \\ 1 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \text{ 所以 } x \notin \text{null}(A)$$

$$A = \begin{bmatrix} 1 & 0 & -2 & 6 \\ -3 & 6 & 6 & -6 \\ 2 & -3 & -4 & 6 \end{bmatrix}, \text{ find } x \in \text{null}(A).$$

$$Ax = 0 \rightarrow A \text{ 做 ref} \rightarrow \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ 0 & 0 & -2 & 6 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 + 0 - 2x_3 + 6x_4 \\ 0 + x_2 + 0 + 2x_4 \\ 0 + 0 + 0 + 0 \\ 0 + 0 + 0 + 0 \end{bmatrix} = 0$$

$$x_1 = 2x_3 - 6x_4$$

$$x_2 = -2x_4$$

$$x_3 = x_3, x_4 = x_4$$

$$\Rightarrow x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2x_3 - 6x_4 \\ -2x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -6 \\ -2 \\ 0 \\ 1 \end{bmatrix}, \left\{\begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -6 \\ -2 \\ 0 \\ 1 \end{bmatrix}\right\} = \text{null}(A)$$

$$A = \begin{bmatrix} 3 & 0 & 3 & 3 & 3 \\ -3 & 1 & -2 & -4 & -1 \\ 5 & 4 & 9 & 1 & 13 \\ 7 & 6 & 13 & 1 & 19 \end{bmatrix} \quad \text{null}(A)? \quad \text{nullity}(A)?$$

$$\xrightarrow{\text{ref}} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ \textcircled{1} & 0 & 1 & 1 & 1 \\ 0 & \textcircled{1} & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -x_3 - x_4 - x_5 \\ -x_3 + x_4 - 2x_5 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{null}(A) \text{Basis} = \left\{ \underset{1}{\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}}, \underset{2}{\begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}}, \underset{3}{\begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}} \right\} \Rightarrow \text{nullity}(A) = 3$$

04/22 Problem

1. $A = \left\{ \begin{bmatrix} 2-b+2c \\ 3+b-2c \\ b+2b+3c \end{bmatrix}, b, c \in \mathbb{R} \right\}$ proof A 是 $\mathbb{R}^3$ 的子空間嗎

$$A = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + c \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} \quad \begin{array}{l} -b+2c=2 \\ b-2c=3 \\ \hline 0 \neq 5 \end{array}$$

$\begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} \notin \text{span} \left(\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} \right)$, 所以 A 不是 $\mathbb{R}^3$ 的 subspace.

2. $A = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 1 & 2 & 3 & 1 \\ -1 & 3 & 2 & 1 \\ -1 & 4 & 3 & 0 \end{bmatrix}$, find $\text{col}(A)$ 的 basis.

$$\text{Span} \left(\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right) = \text{col}(A)$$

$$\begin{bmatrix} 1 & 1 & 2 & 0 \\ 1 & 2 & 3 & 1 \\ -1 & 3 & 2 & 1 \\ -1 & 4 & 3 & 0 \end{bmatrix} \xrightarrow{\substack{r_3 = r_3 + r_1 \\ r_4 = r_4 + r_1}} \begin{bmatrix} 1 & 1 & 2 & 0 \\ 1 & 2 & 3 & 1 \\ 0 & 5 & 5 & 1 \\ 0 & 6 & 6 & 1 \end{bmatrix} \xrightarrow{r_2 = r_2 - r_1} \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 5 & 5 & 1 \\ 0 & 6 & 6 & 1 \end{bmatrix}$$

$$\begin{array}{l} r_3 = r_3 - 5r_2 \\ r_4 = r_4 - 6r_2 \end{array} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & -5 \end{bmatrix} \quad \begin{array}{l} r_3 = -\frac{1}{3}r_3 \\ r_4 = -\frac{1}{5}r_4 \end{array} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{col}(A) \text{ 的 Basis} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right\} \neq$$

$$\xrightarrow{r_4 = r_4 - r_3} \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3. 擴充 $\left\{ \begin{bmatrix} 2 \\ 0 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \\ 3 \\ 0 \end{bmatrix} \right\}$ 為 R^3 的 Basis.

$$\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 4 & 5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{r_3 = r_3 - 2r_1} \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_1 = r_1 + r_2, r_2 = \frac{1}{3}r_2} \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_1 = \frac{1}{2}r_1} \begin{bmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 - r_2} \begin{bmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{r_3 = -\frac{1}{2}r_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \left\{ \begin{bmatrix} 2 \\ 0 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

4. 好 $\text{rank}(A) > \text{nullity}(A)$

$$\text{rank}(A) = \dim(\text{col}(A))$$

$$\text{nullity}(A) = \dim(\text{null}(A))$$

$$\text{rank}(A_{m \times n}) + \text{nullity}(A_{m \times n}) = n$$

5. $A = \begin{bmatrix} 1 & 1 & 5 & 1 \\ 2 & 4 & 14 & 4 \\ 2 & 3 & 12 & 3 \end{bmatrix}$, find $\text{col}(A)$ 和 $\text{null}(A)$ 的 basis, 求 $\dim(\text{col}(A))$ 和

$$\text{nullity}(A)$$

$$\text{col}(A) \text{ 的 Basis} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix} \right\}$$

$$\text{col}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 14 \\ 12 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix} \right\}$$

$$\dim(\text{col}(A)) = R^2$$

$$\begin{bmatrix} 1 & 1 & 5 & 1 \\ 2 & 4 & 14 & 4 \\ 2 & 3 & 12 & 3 \end{bmatrix} \xrightarrow{r_2 = r_2 - r_1, r_3 = r_3 - r_1} \begin{bmatrix} 1 & 1 & 5 & 1 \\ 0 & 3 & 9 & 3 \\ 0 & 2 & 7 & 2 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 - 2r_2} \begin{bmatrix} 1 & 1 & 5 & 1 \\ 0 & 3 & 9 & 3 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow{r_2 = r_2 + r_3} \begin{bmatrix} 1 & 1 & 5 & 1 \\ 0 & 4 & 11 & 3 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow{r_1 = r_1 - r_3} \begin{bmatrix} 1 & 0 & 4 & 1 \\ 0 & 4 & 11 & 3 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow{r_2 = r_2 - 4r_3} \begin{bmatrix} 1 & 0 & 4 & 1 \\ 0 & 0 & 7 & 3 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow{r_2 = \frac{1}{7}r_2} \begin{bmatrix} 1 & 0 & 4 & 1 \\ 0 & 0 & 1 & \frac{3}{7} \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow{r_1 = r_1 - 4r_2} \begin{bmatrix} 1 & 0 & 0 & \frac{13}{7} \\ 0 & 0 & 1 & \frac{3}{7} \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\xrightarrow{r_3 = r_3 - r_2} \begin{bmatrix} 1 & 0 & 0 & \frac{13}{7} \\ 0 & 0 & 1 & \frac{3}{7} \\ 0 & 1 & 0 & -\frac{4}{7} \end{bmatrix}$$

$$\xrightarrow{r_3 = -\frac{4}{7}r_3} \begin{bmatrix} 1 & 0 & 0 & \frac{13}{7} \\ 0 & 0 & 1 & \frac{3}{7} \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

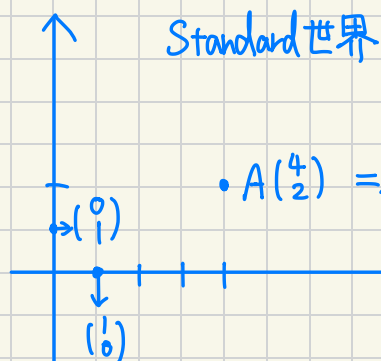
$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ 1 & 0 & -3 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 = 3x_3 \\ x_2 = -2x_3 - x_4 \end{cases}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3x_3 \\ -2x_3 - x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{null}(A) \text{ 的 Basis: } \left\{ \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\text{nullity}(A) = 2$$

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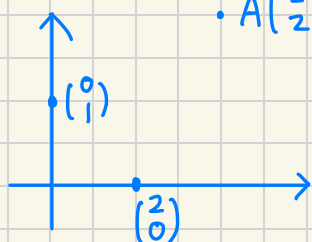


$$\bullet A \begin{pmatrix} 4 \\ 2 \end{pmatrix} \Rightarrow 4 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$\mathbb{R}^2$ 空間的 Basis: $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

β 世界



$$\bullet A \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

有可能因為 Basis 不同,

座標表示也不同

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_S = \begin{bmatrix} 2 \\ 2 \end{bmatrix}_\beta$$

β 世界的 Basis $\left\{ \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

座標變換:

$\beta \rightarrow S$:

$$[P_{\beta \rightarrow S}] [\text{input}]_\beta = [\text{Output}]_S$$

(β) (Standard)

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix}_\beta = \begin{bmatrix} 4 \\ 2 \end{bmatrix}_S$$

↑ ↑
 β_1, β_2
(基底)

$S \rightarrow \beta$:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}_S = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}_\beta$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}_S = \begin{pmatrix} 0 \\ 1 \end{pmatrix}_\beta$$

$$\Rightarrow P_{S \rightarrow \beta} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}$$

↓ ↘
 $[S_1]_\beta$ $[S_2]_\beta$

$$\Rightarrow P_{S \rightarrow \beta} = [S_1]_\beta [S_2]_\beta$$

$$\Rightarrow P_{\beta \rightarrow S} = [\beta_1]_S [\beta_2]_S$$

座標变换 · 求基底?

$$P_{S \rightarrow \beta} = [S_1]_{\beta} [S_2]_{\beta}$$

$$P_{\beta \rightarrow S} = [\beta_1]_S [\beta_2]_S$$

$$\underbrace{[[\beta_1][\beta_2] \dots [\beta_n]]}_{\beta \text{ 的 Basis 們}} \underbrace{[[S_1]_{\beta}]}_{\text{red}} = \underbrace{[S_1]}_{\text{red}}$$

$$\underbrace{[[\beta_1][\beta_2] \dots [\beta_n]]}_{\text{red}} \underbrace{[[S_2]_{\beta}]}_{\text{red}} = \underbrace{[S_2]}_{\text{red}}$$

$$\begin{aligned} & [[\beta_1][\beta_2] \dots [\beta_n] | [S_1][S_2] \dots [S_n]] \\ \Rightarrow & = [I | P_{S \rightarrow \beta}] \end{aligned}$$

$$P_{S \rightarrow \beta} = [\beta | S] \xrightarrow{\text{ref}} [I | P_{S \rightarrow \beta}]$$

$\alpha = \{ \alpha \text{ 的 Basis} \}$

$\beta = \{ \beta \text{ 的 Basis} \}$

$$P_{\alpha \rightarrow \beta} : [\beta | \alpha] \xrightarrow{\text{ref}} [I | P_{\alpha \rightarrow \beta}]$$

(其他表示法: $[T]_{\alpha}^{\beta}$)

$$P_{\beta \rightarrow \alpha} : [\alpha | \beta] \xrightarrow{\text{ref}} [I | P_{\beta \rightarrow \alpha}]$$

(其他表示法: $[T]_{\beta}^{\alpha}$)

$$[P_{\alpha \rightarrow \beta} | I] \rightarrow \left[\begin{array}{cc|cc} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{array} \right]$$

$$\begin{aligned} r_1 &= \frac{1}{2}r_1 \\ r_2 &= \frac{1}{2}r_2 \\ \xrightarrow{\quad} & \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & \frac{1}{2} \end{array} \right] \\ & \underbrace{\hspace{10em}}_{P_{\beta \rightarrow \alpha}} \end{aligned}$$

e.g.

$$\alpha = \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}, \beta = \left\{ \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}$$

$$P_{\alpha \rightarrow \beta} \rightarrow [\beta | \alpha] \xrightarrow{\text{ref}} [I | P_{\alpha \rightarrow \beta}]$$

$$\left[\begin{array}{cc|cc} 2 & 0 & 4 & 0 \\ 0 & 1 & 0 & 2 \end{array} \right] \xrightarrow{r_1 = \frac{1}{2}r_1} \left[\begin{array}{cc|cc} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \end{array} \right]$$

$$P_{\alpha \rightarrow \beta} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$P_{\beta \rightarrow \alpha} = [\alpha | \beta] \xrightarrow{\text{ref}} [I | P_{\beta \rightarrow \alpha}]$$

$$P_{\beta \rightarrow \alpha} []_{\beta} = []_{\alpha}$$

$$\rightarrow \text{想到} \underbrace{P_{\alpha \rightarrow \beta} P_{\beta \rightarrow \alpha}}_I []_{\beta} = P_{\alpha \rightarrow \beta} []_{\alpha}$$

$$\begin{aligned} P_{\alpha \rightarrow \beta} &= [P_{\beta \rightarrow \alpha}]^{-1} = [P_{\beta \rightarrow \alpha} | I] \xrightarrow{\text{ref}} [I | P_{\alpha \rightarrow \beta}] \\ &\leftarrow \text{key} \rightarrow [A | I] \xrightarrow{\text{ref}} [I | A^{-1}] \end{aligned}$$

$$1. \beta = \left\{ \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}, [x]_{\beta} = \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix}, \text{求 } x = [x]_S$$

$$P_{\beta \rightarrow S} [x]_{\beta} = [x]_S$$

$$P_{\beta \rightarrow S} \rightarrow \begin{bmatrix} I & \tilde{P}_{\beta \rightarrow S} \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 1 \\ 2 & 0 & 1 \\ 4 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 8 \end{bmatrix} \underset{\parallel}{=} [x]_S$$

$$2. \beta = \left\{ \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}, x = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \text{求 } [x]_{\beta}?$$

$$P_{\beta \rightarrow S} [x]_{\beta} = [x]_S, P_{S \rightarrow \beta} [x]_S = [x]_{\beta}$$

$$\begin{bmatrix} 2 & -1 & 1 \\ 2 & 0 & 1 \\ 4 & 3 & 1 \end{bmatrix} [x]_{\beta} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 & -1 & 1 & | & 2 \\ 2 & 0 & 1 & | & 3 \\ 4 & 3 & 1 & | & 4 \end{bmatrix} \xrightarrow{\substack{r_2 = r_2 - r_1 \\ r_3 = r_3 - 2r_1}} \begin{bmatrix} 2 & -1 & 1 & | & 2 \\ 0 & 1 & 0 & | & 1 \\ 0 & 5 & -1 & | & 0 \end{bmatrix}$$

$$\xrightarrow{\substack{r_1 = r_1 + r_2 \\ r_3 = r_3 - 5r_2}} \begin{bmatrix} 2 & 0 & 1 & | & 3 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & -1 & | & -5 \end{bmatrix} \xrightarrow{r_3 = -r_3} \begin{bmatrix} 2 & 0 & 1 & | & 3 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & 5 \end{bmatrix} \xrightarrow{r_1 = r_1 - r_3} \begin{bmatrix} 2 & 0 & 0 & | & -2 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & 5 \end{bmatrix}$$

$$\xrightarrow{r_1 = \frac{1}{2}r_1} \begin{bmatrix} 1 & 0 & 0 & | & -1 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & 5 \end{bmatrix}, [x]_{\beta} = \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}$$

$$3. A = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} \right\}, B = \left\{ \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} \right\}, P_{B \rightarrow A}?$$

$$[A|B] = \begin{bmatrix} 1 & 1 & 2 & | & 3 & 0 & -1 \\ 0 & 1 & 2 & | & 1 & 0 & 2 \\ 1 & 0 & -3 & | & 1 & 1 & 0 \end{bmatrix} \xrightarrow{r_3 = r_3 - r_1} \begin{bmatrix} 1 & 1 & 2 & | & 3 & 0 & -1 \\ 0 & 1 & 2 & | & 1 & 0 & 2 \\ 0 & -1 & -5 & | & -2 & 1 & 1 \end{bmatrix} \xrightarrow{\substack{r_1 = r_1 - r_2 \\ r_3 = r_3 + r_2}} \begin{bmatrix} 1 & 0 & 0 & | & 2 & 0 & -3 \\ 0 & 1 & 2 & | & 1 & 0 & 2 \\ 0 & 0 & -3 & | & -1 & 1 & 3 \end{bmatrix}$$

$$\xrightarrow{r_3 = -\frac{1}{3}r_3} \begin{bmatrix} 1 & 0 & 0 & | & 2 & 0 & -3 \\ 0 & 1 & 2 & | & 1 & 0 & 2 \\ 0 & 0 & 1 & | & \frac{1}{3} & -\frac{1}{3} & -1 \end{bmatrix} \xrightarrow{r_2 = r_2 - 2r_3} \begin{bmatrix} 1 & 0 & 0 & | & 2 & 0 & -3 \\ 0 & 1 & 0 & | & \frac{1}{3} & \frac{2}{3} & 4 \\ 0 & 0 & 1 & | & \frac{1}{3} & -\frac{1}{3} & -1 \end{bmatrix}$$

$$\underline{\underline{P_{B \rightarrow A}}}$$

內積

$$(2, 3) \cdot (3, 1) = 2 \times 3 + 3 \times 1 = 9$$

→ 在學中, 內積表示成這樣: $\langle x, y \rangle$

一些性質:

$$\langle x+z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$$

$$\langle cx, y \rangle = c \langle x, y \rangle$$

$$\langle x, y \rangle = \langle y, x \rangle$$

$$\text{if } x \neq 0, \langle x, x \rangle > 0$$

$$\mathbb{R}^n = \begin{matrix} x = (a_1, a_2, a_3, \dots, a_n) \\ y = (b_1, b_2, b_3, \dots, b_n) \end{matrix}$$

$$\langle x, y \rangle = \sum_{i=1}^n a_i b_i$$

— 世界 $\forall [a, b], f \in V, g \in V$

$$\langle f, g \rangle = \int_a^b f(t)g(t)dt$$

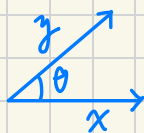
一個向量 x :

$$|x|: x \text{ 的向量長度} = \sqrt{\langle x, x \rangle}$$

單位向量: $|\vec{u}| = 1$ (Unit Vector)

$$\text{正規化: } \frac{\vec{v}}{|\vec{v}|} = \vec{u}$$

$$\text{正射影長: } |\vec{x}||\vec{y}|\cos\theta$$



04/29 Problem

$$\alpha = \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} \right\}, \quad \beta = \left\{ \begin{pmatrix} -3 \\ 2 \\ -3 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} 5 \\ 4 \\ 9 \end{pmatrix} \right\}$$

1. $P_{\alpha \rightarrow \beta}$? $P_{\beta \rightarrow \alpha}$?

$$P_{\beta \rightarrow \alpha} = \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 1 & 0 \\ 1 & -1 & 3 & 0 & 0 & 1 \end{array} \right]$$

$$\begin{array}{l} r_2 = r_2 - 2r_1 \\ r_3 = r_3 - r_1 \end{array} \rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 4 & -1 & -2 & 1 & 0 \\ 0 & 0 & 2 & -1 & 0 & 1 \end{array} \right] \xrightarrow{r_3 = \frac{1}{2}r_3} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 4 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right]$$

$$\xrightarrow{r_2 = r_2 + r_3} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 4 & 0 & -\frac{5}{2} & 1 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right] \xrightarrow{r_2 = \frac{1}{4}r_2} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -\frac{5}{8} & \frac{1}{4} & \frac{1}{8} \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right]$$

$$\xrightarrow{r_1 = r_1 + r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{3}{8} & \frac{1}{4} & \frac{1}{8} \\ 0 & 1 & 0 & -\frac{5}{8} & \frac{1}{4} & \frac{1}{8} \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right]$$

$$\xrightarrow{r_1 = r_1 - r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{8} & \frac{1}{4} & -\frac{3}{8} \\ 0 & 1 & 0 & -\frac{5}{8} & \frac{1}{4} & \frac{1}{8} \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right]$$

$$P_{\alpha \rightarrow \beta} = \left[\begin{array}{ccc} 1 & -1 & 1 \\ 2 & 2 & 1 \\ 1 & -1 & 3 \end{array} \right] \#$$

2. $\checkmark$ $X = \begin{pmatrix} 5 \\ 10 \\ 13 \end{pmatrix}$, $[X]_a$?

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 5 \\ 2 & 2 & 1 & 10 \\ 1 & -1 & 3 & 13 \end{array} \right] \xrightarrow{\substack{r_2=r_2-2r_1 \\ r_3=r_3-r_1}} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 5 \\ 0 & 4 & -1 & 0 \\ 0 & 0 & 2 & 8 \end{array} \right]$$

$$\xrightarrow{r_3=\frac{1}{2}r_3} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 5 \\ 0 & 4 & -1 & 0 \\ 0 & 0 & 1 & 4 \end{array} \right] \xrightarrow{\substack{r_2=r_2+r_3 \\ r_1=r_1-r_3}} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 4 & 0 & 4 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

$$\xrightarrow{r_2=\frac{1}{4}r_2} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 4 \end{array} \right] \xrightarrow{r_1=r_1+r_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 4 \end{array} \right], [X]_a = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$$

3. $\checkmark$ $[X]_a = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $X = ?$

$$\left[\begin{array}{ccc} 1 & -1 & 1 \\ 2 & 2 & 1 \\ 1 & -1 & 3 \end{array} \right] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}, X = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}$$

4. $\checkmark$ $P_A \rightarrow B$? $P_B \rightarrow a$?

$$P_{B \rightarrow A} = \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & -3 & 1 & 5 \\ 2 & 2 & 1 & 2 & -1 & 4 \\ 1 & -1 & 3 & -3 & -1 & 9 \end{array} \right] \xrightarrow{\substack{r_2=r_2-2r_1 \\ r_3=r_3-r_1}} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & -3 & 1 & 5 \\ 0 & 4 & -1 & 8 & -3 & -6 \\ 0 & 0 & 2 & 0 & -2 & 4 \end{array} \right]$$

$$\xrightarrow{r_3=\frac{1}{2}r_3} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & -3 & 1 & 5 \\ 0 & 4 & -1 & 8 & -3 & -6 \\ 0 & 0 & 1 & 0 & -1 & 2 \end{array} \right] \xrightarrow{\substack{r_2=r_2+r_3 \\ r_1=r_1-r_3}} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & -3 & 2 & 3 \\ 0 & 4 & 0 & 8 & -4 & -4 \\ 0 & 0 & 1 & 0 & -1 & 2 \end{array} \right]$$

$$\xrightarrow{r_2=\frac{1}{4}r_2} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & -3 & 2 & 3 \\ 0 & 1 & 0 & 2 & -1 & -1 \\ 0 & 0 & 1 & 0 & -1 & 2 \end{array} \right] \xrightarrow{r_1=r_1+r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 2 \\ 0 & 1 & 0 & 2 & -1 & -1 \\ 0 & 0 & 1 & 0 & -1 & 2 \end{array} \right]$$

$$P_{B \rightarrow A} = \begin{bmatrix} -1 & 1 & 2 \\ 2 & -1 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$P_{A \rightarrow B} \Rightarrow \left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 2 & -1 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{r_2 = r_2 + 2r_1} \left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 2 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{array} \right] \xrightarrow{r_3 = r_3 + r_2} \left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 2 & 1 & 0 \\ 0 & 0 & 5 & 2 & 1 & 1 \end{array} \right]$$

$$\xrightarrow{r_3 = \frac{1}{5}r_3} \left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 2 & 1 & 0 \\ 0 & 0 & 1 & \frac{2}{5} & \frac{1}{5} & \frac{1}{5} \end{array} \right] \xrightarrow{r_2 = r_2 - 3r_3} \left[\begin{array}{ccc|ccc} -1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{8}{5} & \frac{2}{5} & \frac{4}{5} \\ 0 & 0 & 1 & \frac{2}{5} & \frac{1}{5} & \frac{1}{5} \end{array} \right]$$

$$\xrightarrow{r_1 = r_1 - 2r_2} \left[\begin{array}{ccc|ccc} -1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{8}{5} & \frac{2}{5} & \frac{4}{5} \\ 0 & 0 & 1 & \frac{2}{5} & \frac{1}{5} & \frac{1}{5} \end{array} \right] \xrightarrow{r_1 = r_1 - r_2} \left[\begin{array}{ccc|ccc} -1 & 0 & 0 & -\frac{3}{5} & -\frac{2}{5} & -\frac{3}{5} \\ 0 & 1 & 0 & \frac{8}{5} & \frac{2}{5} & \frac{4}{5} \\ 0 & 0 & 1 & \frac{2}{5} & \frac{1}{5} & \frac{1}{5} \end{array} \right]$$

$$P_{A \rightarrow B} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{3}{5} & -\frac{2}{5} & -\frac{3}{5} \\ 0 & 1 & 0 & \frac{8}{5} & \frac{2}{5} & \frac{4}{5} \\ 0 & 0 & 1 & \frac{2}{5} & \frac{1}{5} & \frac{1}{5} \end{array} \right]$$

5. $[X]_A = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}, [X]_B = ?$

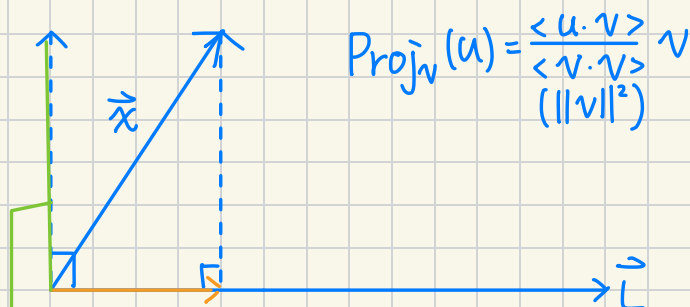
$$\frac{6}{5} + \frac{2}{5}$$

$$\left[\begin{array}{ccc} \frac{3}{5} & \frac{4}{5} & -\frac{1}{5} \\ \frac{4}{5} & \frac{2}{5} & -\frac{3}{5} \\ \frac{2}{5} & -\frac{1}{5} & \frac{1}{5} \end{array} \right] \left[\begin{array}{c} 3 \\ 0 \\ 2 \end{array} \right] = \left[\begin{array}{c} \frac{1}{5} \\ \frac{10}{5} \\ \frac{1}{5} \end{array} \right] \neq$$

$$[X]_B = \left[\begin{array}{c} \frac{1}{5} \\ \frac{10}{5} \\ \frac{1}{5} \end{array} \right] \neq$$

05 / 06

Gram-Schmidt 格拉姆-施密特正交化 (單範正交化)



x 投影在 L 上的向量: $\text{Proj}_L(\vec{x})$

$$\vec{x} - \text{Proj}_L(\vec{x})$$

$$\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 8 \\ 1 \\ -6 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$x_1 \quad x_2 \quad x_3$

做 Gram-Schmidt 得一組 Orthogonal Basis.
(正交基)
(標準正交基)
一組 Orthonormal Basis.

$$v_1 = x_1$$

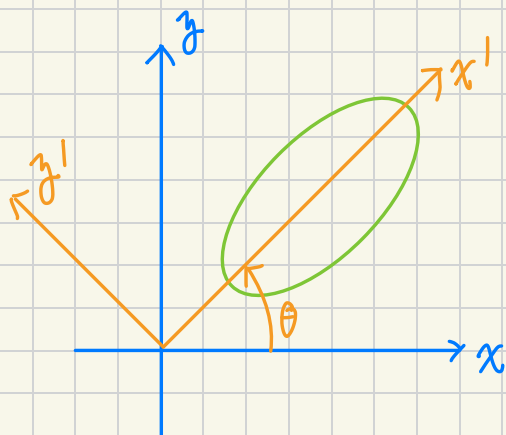
$$v_2 = x_2 - \text{Proj}_{v_1}(x_2) = \begin{bmatrix} 8 \\ 1 \\ -6 \end{bmatrix} - \frac{\langle \vec{v}_1, \vec{x}_2 \rangle}{\|\vec{v}_1\|^2} \vec{v}_1 = \begin{bmatrix} 8 \\ 1 \\ -6 \end{bmatrix} - \frac{10}{5} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 6 \end{bmatrix}$$

$$v_3 = x_3 - \text{Proj}_{v_1}(x_3) - \text{Proj}_{v_2}(x_3) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - \frac{0}{5} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \frac{-6}{81} \begin{bmatrix} 6 \\ -3 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{4}{9} \\ \frac{2}{9} \\ \frac{5}{9} \end{bmatrix}$$

$$(1) \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -3 \\ 6 \end{bmatrix}, \begin{bmatrix} \frac{4}{9} \\ \frac{2}{9} \\ \frac{5}{9} \end{bmatrix} \right\} \quad (2) \left\{ \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \frac{1}{3} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \frac{1}{\sqrt{5}} \begin{bmatrix} 4 \\ -2 \\ -5 \end{bmatrix} \right\}$$

除長度, 就是標準正交基

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$



$$\cot 2\theta = \frac{a-c}{b}$$

$$a+c = a'+c'$$

$$b^2 - 4ac = (b')^2 - 4a'c'$$

原理.....

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$x = x' \cos\theta - y' \sin\theta$$

$$y = x' \sin\theta + y' \cos\theta$$

$$a' = a \cos^2\theta + b \sin\theta \cos\theta + c \sin^2\theta$$

求 $a'x'^2 + b'x'y' + c'y'^2 + d'x' + e'y' + f' = 0 \rightarrow$ 看成正的橢圓!

橢圓是沒有這項的, $\therefore b' = 0$

e.g.

$$5x^2 + \sqrt{3}xy + 4y^2 + 3x - y + 10 = 0, \text{ 轉成正橢圓}$$

$$\cot 2\theta = \frac{5-4}{\sqrt{3}} = \frac{1}{\sqrt{3}}, \quad 2\theta = 60^\circ, \quad \theta = 30^\circ \quad (\cot 2\theta = \frac{1}{\tan 2\theta})$$

$$a' = 5 \times \frac{3}{4} + \sqrt{3} \times \frac{1}{2} \times \frac{\sqrt{3}}{2} + 4 \times \frac{1}{4} = \frac{15}{4} + \frac{3}{4} + 1 = \frac{11}{2}$$

$$c' = 5 + 4 - \frac{11}{2} = \frac{7}{2}$$

$$(a+c = a'+c')$$

驗證: $b^2 - 4ac$, b' 必為 0

$$\sqrt{3}^2 - 4 \times 5 \times 4 = b'^2 - 4 \times \frac{11}{2} \times \frac{7}{2}$$

$$-77 = b'^2 - 77, \quad b' = 0 \quad \checkmark$$

係數 $d, e \dots$

$$d' = d \cos \theta + e \sin \theta = 3 \times \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{3\sqrt{3}}{2} - \frac{1}{2}$$

$$e' = -d \sin \theta + e \cos \theta = -3 \times \frac{1}{2} - \frac{\sqrt{3}}{2} = -\frac{3}{2} - \frac{\sqrt{3}}{2}$$

$$f = f', \quad f' = 10$$

$$\Rightarrow \frac{1}{2}x^2 + \frac{1}{2}y^2 + \left(\frac{3\sqrt{3}}{2} - \frac{1}{2}\right)x + \left(-\frac{3}{2} - \frac{\sqrt{3}}{2}\right)y + 10 = 0$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}, \quad \begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= -x \sin \theta + y \cos \theta \end{aligned}$$

→ 新的 x, y 關係 $\dots$

$$x' = \frac{\sqrt{3}}{2}x + \frac{1}{2}y$$

$$y' = -\frac{1}{2}x + \frac{\sqrt{3}}{2}y$$

<

公告

參考習題

發布時間: 2025-05-01 14:08:47

4.1 習題: 15, 21, 47, 51

4.3 習題: 1, 2, 13, 16, 20, 39, 41, 44

4.4 習題: 1, 33, 35, 38, 61, 62

4.5 習題: 1, 2, 7, 8, 9, 40, 41, 43, 45, 46, 69, 70, 73, 74

4.6 習題: 3, 4, 11, 12, 27, 33, 41, 42, 45, 55, 59, 61, 68, 69

4.7 習題: 3, 9, 15, 19, 23, 37, 39

4.8 習題: 61, 63, 65, 67, 69, 73, 75, 77

5.1 習題: 1, 3, 5, 10, 12, 16, 21, 23, 83, 84

5.2 習題: 2, 4, 9, 13, 15, 17, 23, 48, 73~76, 90

5.3 習題: ~~1, 5, 13, 15~~, 23, 25, 29, 33, 37, 39

5.4 習題: ~~23, 31, 37~~

轉置性質 (課 P.81)

$$1. (AB)^T = B^T A^T$$

2. $(AB)^T$ 的第 (i, j) 元素
= AB 的第 (j, i) 元素

3. $A_{m \times n}$ 可逆, 則 A^T 亦可逆

$$4. (A^T)^T = A$$

$$A^T A x = A^T b$$

$$x = (A^T A)^{-1} A^T b$$

05/06 Problem

1. 將 $\begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$ 做 Gram-Schmidt 得標準正交基
 $x_1 \quad x_2 \quad x_3$

$$v_1 = x_1$$

$$v_2 = x_2 - \text{Proj}_{v_1}(x_2) = \begin{bmatrix} 5 \\ 1 \\ 3 \end{bmatrix} - \frac{18}{18} \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix}$$

$$\begin{aligned} v_3 &= x_3 - \text{Proj}_{v_1}(x_3) - \text{Proj}_{v_2}(x_3) = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} - \frac{8}{18} \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} - \frac{0}{17} \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{6} \\ \frac{1}{6} \\ \frac{2}{3} \end{bmatrix} = \begin{bmatrix} -\frac{1}{6} \\ \frac{17}{6} \\ -\frac{2}{3} \end{bmatrix} \end{aligned}$$

$$\text{正交基: } \left\{ \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \\ 1 \end{bmatrix} \right\}$$

$$\text{標準正交基: } \left\{ \frac{1}{\sqrt{18}} \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}, \frac{1}{\sqrt{17}} \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix}, \frac{1}{\sqrt{\frac{17}{2}}} \begin{bmatrix} -\frac{1}{6} \\ \frac{17}{6} \\ -\frac{2}{3} \end{bmatrix} \right\}$$

2. 有一斜橢圓 $6x^2 + 3\sqrt{3}xy + 3y^2 - 2x - y + 10 = 0$, 轉正。

$$\cot 2\theta = \frac{a-c}{b} = \frac{6-3}{3\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}, \quad 2\theta = 60^\circ, \quad \theta = 30^\circ$$

$$a' = 6 \times \frac{3}{4} + 3\sqrt{3} \times \frac{1}{2} \times \frac{\sqrt{3}}{2} + 3 \times \frac{1}{4} = \frac{9}{2} + \frac{9}{4} + \frac{3}{4} = \frac{18}{4} + \frac{12}{4} = \frac{30}{4} = \frac{15}{2} \neq$$

$$a+c = a'+c', \quad c' = 6+3 - \frac{15}{2} = \frac{18}{2} - \frac{15}{2} = \frac{3}{2} \neq$$

$$d = -2 \times \frac{\sqrt{3}}{2} - \frac{1}{2} = (-\sqrt{3} - \frac{1}{2})$$

$$e = 1 - \frac{\sqrt{3}}{2}$$

$$\frac{15}{2}x'^2 + \frac{3}{2}y'^2 + (-\sqrt{3} - \frac{1}{2})x' + (1 - \frac{\sqrt{3}}{2})y' + 10 = 0$$

$$x' = \frac{\sqrt{3}}{2}x + \frac{1}{2}y$$

$$y' = -\frac{1}{2}x + \frac{\sqrt{3}}{2}y$$

QR 分解

QR 分解

QR 分解就是將一個矩陣 A 分解成 QR ，就是 $A = QR$ 。

其中 Q 是由 orthonormal vector 組成的矩陣， R 是上三角矩陣。

性質：

1. $QQ^T = Q^TQ = I$ ，因為不同 orthonormal vector 互相內積為 0，自己與自己內積為 1（因為是單位向量）。

2. $A = QR \rightarrow Q^TA = Q^TQR = R$ 。

題目要你做 QR 分解，就是要你把 $Q = ?$ 、 $R = ?$ 寫出來。

1. → 將 $\begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}$ 做 QR 分解。

$$\text{令 } x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}, x_3 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

經由 gram-schmidt 後得到 orthonormal vector（記得要標準化）。

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ 則 } Q = [u_1 \ u_2 \ u_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\text{則 } R = Q^TA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 3 & 6 \\ 0 & 0 & 5 \end{bmatrix}$$

2. → 將 $\begin{bmatrix} 2 & 3 \\ 2 & 4 \\ 1 & 1 \end{bmatrix}$ 做 QR 分解。

$$\text{令 } x_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

經由 gram-schmidt 後得到 orthonormal vector（記得要標準化）。

$$u_1 = \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}, u_2 = \begin{bmatrix} -\frac{1}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}, \text{ 則 } Q = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{2}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{bmatrix}$$

$$\text{則 } R = Q^TA = \begin{bmatrix} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 2 & 4 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 0 & 1 \end{bmatrix}$$

↓

↓

↓

↓

Q 的大小會跟原本的矩陣一樣。

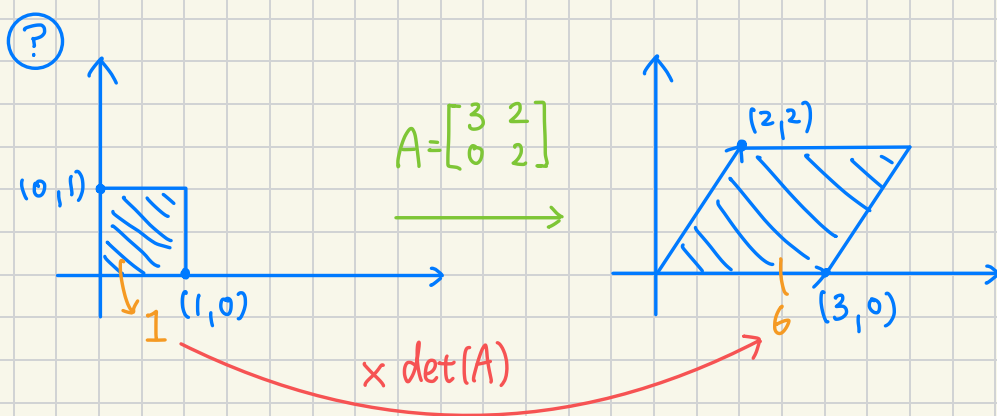
$$R \text{ 應該要長得像 } \begin{bmatrix} X & X & X & \cdots & X \\ 0 & X & X & \cdots & X \\ 0 & 0 & X & \cdots & X \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & X \end{bmatrix}, X \text{ 代表任意數字}$$

驗算答案時，可以直接算 QR 有沒有等於 A 。

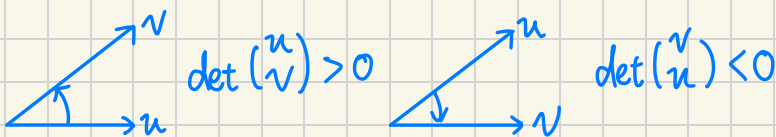
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行列式

- $\det()$: Basis 所繞出體(面)積的變化量 ?
- 反矩陣: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$



方向 Orientation



奇異矩陣 Singular Matrix

— 矩陣 A ...

若 A^{-1} Not Exist, 且 $\det(A) = 0$

則 A 是 Singular Matrix, 反之則不是。

行列式的计算

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$A: \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$B: \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = (aei + dhc + gbf) - (ceg + ahf + bdi)$$

(Ext)

Another Way to Solved This.

ex. $A = \begin{bmatrix} 4 & 5 & 3 \\ 4 & 5 & 3 \\ 3 & 5 & 2 \end{bmatrix}, \det(A) = \begin{vmatrix} 4 & 5 & 3 \\ 4 & 5 & 3 \\ 3 & 5 & 2 \end{vmatrix}$

利用性质:

(1) $\det(A) = \det(A^T), \begin{vmatrix} 4 & 4 & 3 \\ 5 & 5 & 5 \\ 3 & 3 & 2 \end{vmatrix} = \det(A)$

(2) Swap $(r_1, r_2), \begin{vmatrix} 5 & 5 & 5 \\ 4 & 4 & 3 \\ 3 & 3 & 2 \end{vmatrix} = \det(A)$

(3) $\times k, 5 \begin{vmatrix} 1 & 1 & 1 \\ 4 & 4 & 3 \\ 3 & 3 & 2 \end{vmatrix} = \det(A)$

(4) $r_2 = r_2 - r_1, -5 \begin{vmatrix} 1 & 1 & 1 \\ 3 & 3 & 2 \\ 3 & 3 & 2 \end{vmatrix} = \det(A)$

(5) 2 行列 Same?, $\det(A) = 0$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

行列式性质

- $\det(A) = \det(A^T)$

- 若 A 的两列 or 两行 互换: $\det(A) * = -1$

- A 的某列 乘上 k: $\det(A) * = k$

- A 的某一列 加上其他列的倍数: $\det(A)$ 不变

- 有全 0 的行 or 列 或有 2 行 or 2 列 一样: $\det(A) = 0$

降阶 $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$ (3x3 正负对应表)

$\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} - a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$

$\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$ (4x4 正负对应表)

ex. $A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 7 & 3 \\ 8 & 9 & 5 \end{bmatrix}$

利用降階:

$$-\det(A) = \begin{vmatrix} 1 & 6 & 4 \\ 2 & 7 & 3 \\ 8 & 9 & 5 \end{vmatrix} = +1 \begin{vmatrix} 7 & 3 \\ 9 & 5 \end{vmatrix} - 2 \begin{vmatrix} 6 & 4 \\ 9 & 5 \end{vmatrix} + 8 \begin{vmatrix} 6 & 4 \\ 7 & 3 \end{vmatrix} = -60$$

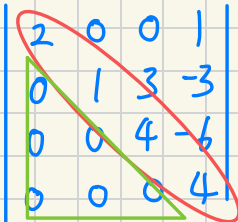
先用性質再降階:

$$\begin{vmatrix} 1 & 6 & 4 \\ 2 & 7 & 3 \\ 8 & 9 & 5 \end{vmatrix} \xrightarrow{\substack{r_2 = r_2 - 2r_1 \\ r_3 = r_3 - 8r_1}} \begin{vmatrix} 1 & 6 & 4 \\ 0 & -5 & -5 \\ 0 & -39 & -27 \end{vmatrix} = 1 \times \begin{vmatrix} -5 & -5 \\ -39 & -27 \end{vmatrix} = -60$$

ex.

$$A = \begin{bmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & -3 \\ -2 & -3 & -5 & 2 \\ 4 & -4 & -4 & -6 \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & -3 \\ -2 & -3 & -5 & 2 \\ 4 & -4 & -4 & -6 \end{vmatrix}$$

many Steps $\rightarrow$ 

$$= 2 \times 1 \times 4 \times 4 = 32$$

用行列式求 3×3 反矩陣

$$A^{-1} = \frac{1}{\det(A)} C^T$$

C 的正負對應表

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \quad C = \begin{bmatrix} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} & \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}$$

ex. $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

$$\det(A) = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{vmatrix} = 1 \times \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - 0 \begin{vmatrix} 0 & 0 \\ 2 & 1 \end{vmatrix} + 1 \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} = -1$$

$$C_{11} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \quad C_{12} = -\begin{vmatrix} 0 & 0 \\ 2 & 1 \end{vmatrix} = 0 \quad C_{13} = \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} = -2$$

$$C_{21} = -\begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix} = 0 \quad C_{22} = \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -1 \quad C_{23} = -\begin{vmatrix} 1 & 0 \\ 2 & 0 \end{vmatrix} = 0$$

$$C_{31} = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1 \quad C_{32} = -\begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} = 0 \quad C_{33} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$$C = \begin{bmatrix} 1 & 0 & -2 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \quad C^T = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \quad A^{-1} = \frac{1}{-1} \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix}$$

05 / 27 Problem

1. $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$, $\det(A)$?

$$\begin{array}{l} r_2 = r_2 - r_1 \\ r_3 = r_3 - r_1 \\ r_4 = r_4 - r_1 \end{array} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 5 & 9 \\ 0 & 3 & 9 & 19 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 9 \\ 3 & 9 & 19 \end{bmatrix} \begin{array}{l} r_2 = r_2 - 2r_1 \\ r_3 = r_3 - 3r_1 \end{array} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 3 \\ 0 & 3 & 10 \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 3 \\ 0 & 3 & 10 \end{vmatrix} = (10 + 0 + 0) - (0 + 9 + 0) = 1 \neq$$

2. $A = \begin{bmatrix} 5 & 2 & 0 & 0 & -2 \\ 0 & 1 & 4 & 3 & 2 \\ 0 & 0 & 2 & 6 & 3 \\ 0 & 0 & 3 & 4 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$, $\det(A)$?

$$\rightarrow 5 \begin{bmatrix} 1 & 4 & 3 & 2 \\ 0 & 2 & 6 & 3 \\ 0 & 3 & 4 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix} \rightarrow 5 \begin{bmatrix} 2 & 6 & 3 \\ 3 & 4 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

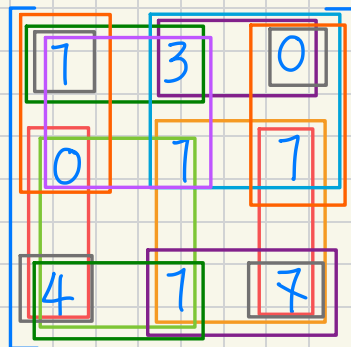
$$\det(A) = 5 \begin{vmatrix} 2 & 6 & 3 \\ 3 & 4 & 1 \\ 0 & 0 & 2 \end{vmatrix} = 5 \times [(16 + 0 + 0) - (0 + 0 + 36)] = -100 \neq$$

$$3. A = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 7 & 7 \\ 4 & 1 & 7 \end{bmatrix}, \det(A) ?$$

$$\det(A) = (7+0+12) - (0+7+0) = 18$$

$$4. A = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 7 & 7 \\ 4 & 1 & 7 \end{bmatrix}, \text{用行列式求 } A^{-1}$$

$$< A^{-1} = \frac{1}{\det(A)} C^T >, C \text{ 正负对应表 } \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$



$$C_{11} = \begin{vmatrix} 7 & 7 \\ 1 & 7 \end{vmatrix} = 6 \quad C_{12} = - \begin{vmatrix} 0 & 7 \\ 4 & 7 \end{vmatrix} = 4 \quad C_{13} = \begin{vmatrix} 0 & 1 \\ 4 & 1 \end{vmatrix} = -4$$

$$C_{21} = - \begin{vmatrix} 3 & 0 \\ 1 & 7 \end{vmatrix} = -2 \quad C_{22} = \begin{vmatrix} 1 & 0 \\ 4 & 7 \end{vmatrix} = 7 \quad C_{23} = - \begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix} = 11$$

$$C_{31} = \begin{vmatrix} 3 & 0 \\ 7 & 7 \end{vmatrix} = 3 \quad C_{32} = - \begin{vmatrix} 1 & 0 \\ 0 & 7 \end{vmatrix} = -1 \quad C_{33} = \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1$$

$$C = \begin{bmatrix} 6 & 4 & -4 \\ -2 & 7 & 11 \\ 3 & -1 & 1 \end{bmatrix}, \quad C^T = \begin{bmatrix} 6 & -2 & 3 \\ 4 & 7 & -1 \\ -4 & 11 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{18} \begin{bmatrix} 6 & -2 & 3 \\ 4 & 7 & -1 \\ -4 & 11 & 1 \end{bmatrix}$$

06 / 10

特徵值(λ)、特徵向量(E)

$$A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}, \quad AE = \lambda E$$

$$\det(\lambda I - A) = 0$$

$$\det(A - \lambda I) = 0$$

求 λ ?

$$\det(\lambda I - A) = 0$$

$$\begin{vmatrix} \lambda - 1 & -1 \\ -4 & \lambda - 1 \end{vmatrix}$$

$$= (\lambda - 1)^2 - 4 = \lambda^2 - 2\lambda - 3$$
$$= (\lambda - 3)(\lambda + 1)$$

$$\lambda = 3, -1$$

找特徵向量 E ?

$$\lambda = 3:$$

$$(\lambda I - A)E = 0$$

$$\begin{bmatrix} 3-1 & -1 \\ -4 & 3-1 \end{bmatrix} E = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x - y = 0$$

$$-4x + 2y = 0$$

$$x = 1, y = 2 \rightarrow E = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\lambda = -1:$$

$$\begin{bmatrix} -1-1 & -1 \\ -4 & -1-1 \end{bmatrix} E = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 & -1 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x - y = 0$$

$$-4x - 2y = 0$$

$$x = 1, y = -2 \rightarrow E = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Another Sample:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \quad \det(\lambda I - A) = 0$$

$$\begin{vmatrix} \lambda - 1 & 0 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 3 \end{vmatrix} = 0 \Rightarrow (\lambda - 1)(\lambda - 2)(\lambda - 3) = 0$$

對角化

$$A = PDP^{-1} \quad / \quad D = P^{-1}AP$$

D 對角矩陣?

只有「對角線」有數字, 其他都是 0

e.g. $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ 即一對角矩陣。

<sol>

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{bmatrix}, \quad \lambda = \underline{2}, \underline{-2}, \underline{3} \quad E = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow P = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 4 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} \underline{2} & 0 & 0 \\ 0 & \underline{-2} & 0 \\ 0 & 0 & \underline{3} \end{bmatrix} \quad \triangle D, E, \lambda \text{ 順序一定要對! , 一題可以有多种答案。}$$

遞迴式轉一般式

$x_n = 2x_{n-1} + 3x_{n-2}$, 求 x_n 一般式? (已知 $x_1=0, x_2=2$)

<sol>

$$\begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_{n-1} \\ x_{n-2} \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}^{n-2} \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix} \text{ 的 } \lambda: 3, -1; D = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}; E = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}; P = \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix} = A^{n-2} \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = (\cancel{PDP^{-1}})(\cancel{PDP^{-1}}) \dots \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} = PD^{n-2}P^{-1} \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}^{n-2} \frac{1}{4} \begin{bmatrix} -1 & -1 \\ -1 & 3 \end{bmatrix} \times \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad < D^n = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}^n = \begin{bmatrix} a^n & 0 \\ 0 & b^n \end{bmatrix}$$

$$= -\frac{1}{4} \times \begin{bmatrix} -2 \times 3^{n-1} & -2 \times (-1)^{n-2} \\ \vdots & \vdots \end{bmatrix} P^{-1} \Rightarrow x_n = \frac{3^n}{6} + \frac{1}{2} \times (-1)^n$$

06 / 10 Problem

1. $A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix}$, 求所有特徵值和其特徵向量。

$$\begin{vmatrix} \lambda-4 & -1 & 0 \\ -1 & \lambda-4 & -1 \\ 0 & -1 & \lambda-4 \end{vmatrix} = \lambda^3 - 12\lambda^2 + 46\lambda - 56$$

$$\begin{vmatrix} \lambda-4 & -1 & 0 \\ -1 & \lambda-4 & -1 \\ 0 & -1 & \lambda-4 \end{vmatrix} = \lambda^3 - 4\lambda^2 - 8\lambda^2 + 32\lambda + 14\lambda - 56$$

$$\lambda^3 - 4\lambda^2 - 8\lambda^2 + 32\lambda + 14\lambda - 56 = \lambda^2(\lambda-4) - 8\lambda(\lambda-4) + 14(\lambda-4)$$

$$= (\lambda-4)(\lambda^2 - 8\lambda + 14), \lambda = 4 \text{ or } 4 \pm \sqrt{2}$$

$$\lambda I - A$$

$$= \begin{bmatrix} \lambda-4 & -1 & 0 \\ -1 & \lambda-4 & -1 \\ 0 & -1 & \lambda-4 \end{bmatrix}$$

$$(\lambda I - A)E = 0$$

$$\lambda = 4 + \sqrt{2}$$

$$\begin{bmatrix} \sqrt{2} & -1 & 0 \\ -1 & \sqrt{2} & -1 \\ 0 & -1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\sqrt{2}x - y = 0, x = \sqrt{2}, y = 2$$

$$-x + \sqrt{2}y - z = 0, z = \sqrt{2}$$

$$-y + \sqrt{2}z = 0$$

$$E_2 = \begin{bmatrix} \sqrt{2} \\ 2 \\ \sqrt{2} \end{bmatrix}$$

$$\lambda = 4 - \sqrt{2}$$

$$\begin{bmatrix} -\sqrt{2} & -1 & 0 \\ -1 & -\sqrt{2} & -1 \\ 0 & -1 & -\sqrt{2} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$-\sqrt{2}x - y = 0$$

$$-x - \sqrt{2}y - z = 0$$

$$-y - \sqrt{2}z = 0$$

$$x = -\sqrt{2}, y = 2$$

$$z = -\sqrt{2}$$

$$E_3 = \begin{bmatrix} -\sqrt{2} \\ 2 \\ -\sqrt{2} \end{bmatrix}$$

$$\lambda = 4$$

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$-y = 0, y = 0$$

$$-x - z = 0$$

$$-y = 0$$

$$x = -1, z = 1$$

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = E_1$$

2, $x_n = 5x_{n-1} - 6x_{n-2}$, $x_1=2$, $x_2=5$, x_n -一般式?

$$\begin{bmatrix} 5 & -6 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_{n-1} \\ x_{n-2} \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & -6 \\ 1 & 0 \end{bmatrix}^{n-2} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \end{bmatrix}$$

$$\langle PD^{n-2}P^{-1} \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} \rangle$$

$$\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{n-2} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3^{n-1} & 2^{n-1} \\ 3^{n-2} & 2^{n-2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_n = 3^{n-1} + 2^{n-1} \quad *$$

$$\begin{vmatrix} \lambda - 5 & 6 \\ -1 & \lambda \end{vmatrix} = \lambda^2 - 5\lambda + 6 \\ = (\lambda - 3)(\lambda - 2), \lambda = 3 \text{ or } 2$$

$$\lambda = 3$$

$$\begin{bmatrix} -2 & 6 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \quad \begin{array}{l} -x + 3y = 0 \\ -2x + 6y = 0 \end{array}$$

$$E_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$x = 3, y = 1$$

$$\lambda = 2:$$

$$\begin{bmatrix} -3 & 6 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \quad \begin{array}{l} -3x + 6y = 0 \\ -x + 2y = 0 \end{array}$$

$$x = 2, y = 1$$

$$E_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$E = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, P = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

